Matemàtiques
1r ESO

SOLUCIONS DE LA FITXA · UNITAT 8

Àrees i perímetres

FentMates

Intenta resoldre cada exercici abans de mirar-ne la solució. Si no t'ix el mateix resultat, repassa els passos o consulta els apunts interactius de la unitat: hi trobaràs eines per a canviar d'unitats i per a explorar les àrees i els perímetres de totes les figures.

1 Perímetre i àrea. Unitats de mesura

1. ★☆☆

a) $A = \boxed{6\ \text{cm}^2}$, $P = \boxed{10\ \text{cm}}$ **b)** $A = \boxed{7\ \text{cm}^2}$, $P = \boxed{16\ \text{cm}}$ **c)** $A = \boxed{5\ \text{cm}^2}$, $P = \boxed{12\ \text{cm}}$

2. ★☆☆

a) $\boxed{4700\ \text{m}}$ **b)** $\boxed{3{,}8\ \text{m}}$ **c)** $\boxed{2{,}5\ \text{m}}$ **d)** $\boxed{900\ \text{m}}$ **e)** $\boxed{1{,}25\ \text{m}}$

3. ★☆☆

a) $\boxed{320\ \text{cm}}$ **b)** $\boxed{4{,}5\ \text{cm}}$ **c)** $\boxed{80\ \text{dam}}$ **d)** $\boxed{0{,}76\ \text{hm}}$

4. ★☆☆

a) $\boxed{500\ \text{cm}^2}$ **b)** $\boxed{240\ \text{dm}^2}$ **c)** $\boxed{3{,}5\ \text{m}^2}$

d) $\boxed{7000\ \text{m}^2}$ **e)** $\boxed{800\ \text{hm}^2}$

5. ★★☆

a) $\boxed{35\,000\ \text{m}^2}$ **b)** $\boxed{0{,}45\ \text{ha}}$ **c)** $\boxed{1200\ \text{m}^2}$ **d)** $\boxed{25\ \text{ha}}$

6. ★★☆ Ho passem tot a m^2: $50\,000$, $40\,000$, $52\,000$ i $48\,000$. $\boxed{4\ \text{ha} < 48\,000\ \text{m}^2 < 0{,}05\ \text{km}^2 < 520\ \text{dam}^2}$

7. ★★☆ $2{,}4\ \text{ha} = 24\,000\ \text{m}^2$; $24\,000 \cdot 18 = \boxed{432\,000\ €}$

8. ★★★ La base i l'altura sumen 10: 1×9 ($A = 9$), 2×8 ($A = 16$), 3×7 ($A = 21$), 4×6 ($A = 24$) i 5×5 ($A = 25$). L'àrea més gran és la del $\boxed{\text{quadrat de 5 cm}}$: $25\ \text{cm}^2$.

CC BY-NC-SA 4.0 · fentmates.org

2 Perímetre dels polígons

9. ★☆☆

a) $\boxed{27{,}5\text{ cm}}$

b) $4 \cdot 8{,}5 = \boxed{34\text{ cm}}$

c) $2 \cdot 15 + 2 \cdot 9 = \boxed{48\text{ m}}$

d) $4 \cdot 6{,}3 = \boxed{25{,}2\text{ cm}}$

10. ★☆☆

a) $5 \cdot 7 = \boxed{35\text{ cm}}$

b) $6 \cdot 4{,}5 = \boxed{27\text{ cm}}$

c) $8 \cdot 3{,}2 = \boxed{25{,}6\text{ cm}}$

d) $10 \cdot 12 = \boxed{120\text{ mm}}$

11. ★☆☆ $P = 5 + 3 + 4 + 3{,}5 + 3 = \boxed{18{,}5\text{ cm}}$

12. ★★☆

a) $46 : 4 = \boxed{11{,}5\text{ cm}}$

b) $51 : 6 = \boxed{8{,}5\text{ cm}}$

c) $13{,}5 : 3 = \boxed{4{,}5\text{ m}}$

13. ★★☆ Les dues bases fan 40 cm; queden $64 - 40 = 24$ cm per a les dues altures: $h = 24 : 2 = \boxed{12\text{ cm}}$.

14. ★★☆ $P = 2 \cdot 120 + 2 \cdot 85 = 410$ m. Amb tres fils: $3 \cdot 410 = \boxed{1230\text{ m}}$. Preu: $1230 \cdot 0{,}35 = \boxed{430{,}50\text{ €}}$.

15. ★★★ El perímetre equival a $3 + 1 + 3 + 1 = 8$ altures: $h = 96 : 8 = \boxed{12\text{ cm}}$ i $b = 3 \cdot 12 = \boxed{36\text{ cm}}$.

16. ★★★ El perímetre equival a $1 + 2 + 2 = 5$ bases: base $= 40 : 5 = \boxed{8\text{ cm}}$ i els costats iguals, $\boxed{16\text{ cm}}$ cadascun.

3 Àrea del quadrat i del rectangle

17. ★☆☆

a) $A = \boxed{144\text{ cm}^2}$, $P = \boxed{48\text{ cm}}$

b) $A = \boxed{120\text{ cm}^2}$, $P = \boxed{46\text{ cm}}$

c) $A = \boxed{30\text{ m}^2}$, $P = \boxed{23\text{ m}}$

d) $A = \boxed{6{,}25\text{ m}^2}$, $P = \boxed{10\text{ m}}$

18. ★☆☆ $h = 84 : 12 = \boxed{7\text{ cm}}$; $P = 2 \cdot 12 + 2 \cdot 7 = \boxed{38\text{ cm}}$

19. ★☆☆

a) $11^2 = 121 \Rightarrow l = \boxed{11\text{ m}}$; $P = \boxed{44\text{ m}}$

b) $l = 28 : 4 = 7$ cm; $A = 7^2 = \boxed{49\text{ cm}^2}$

20. ★★☆ $A = 5{,}4 \cdot 3{,}5 = 18{,}9\text{ m}^2$; $18{,}9 \cdot 24 = \boxed{453{,}60\text{ €}}$

Llicència Creative Commons Reconeixement-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

21. ★★☆ $6 \cdot 2{,}5 - 1{,}5 \cdot 1{,}2 = 15 - 1{,}8 = 13{,}2$ m^2. $13{,}2 : 5 = 2{,}64$: cal comprar $\boxed{3 \text{ pots}}$.

22. ★★☆ Caben $800 : 50 = 16$ rajoles al llarg i $600 : 50 = 12$ a l'ample: $16 \cdot 12 = \boxed{192}$ rajoles.

23. ★★★ P: de 20 cm a 40 cm, $\boxed{\text{es duplica}}$. A: de 25 cm^2 a 100 cm^2, $\boxed{\text{es multiplica per 4}}$.

24. ★★★ 1×36 ($P = 74$), 2×18 ($P = 40$), 3×12 ($P = 30$), 4×9 ($P = 26$), 6×6 ($P = 24$). El de perímetre més menut és el $\boxed{\text{quadrat de 6 cm}}$.

4 Àrea del romboide i del triangle

25. ★☆☆

a) $12 \cdot 7 = \boxed{84 \text{ cm}^2}$ **b)** $8{,}5 \cdot 4 = \boxed{34 \text{ m}^2}$

26. ★☆☆

a) $\frac{14 \cdot 9}{2} = \boxed{63 \text{ cm}^2}$ **b)** $\frac{7 \cdot 5}{2} = \boxed{17{,}5 \text{ cm}^2}$ **c)** $\frac{6{,}4 \cdot 3{,}5}{2} = \boxed{11{,}2 \text{ m}^2}$

27. ★☆☆

a) $\frac{6 \cdot 4}{2} = \boxed{12 \text{ cm}^2}$

b) $\frac{8 \cdot 5}{2} = \boxed{20 \text{ cm}^2}$

c) $\frac{4 \cdot 5}{2} = \boxed{10 \text{ cm}^2}$ (l'altura cau fora)

28. ★★☆ Els catets fan de base i altura: $A = \frac{9 \cdot 12}{2} = \boxed{54 \text{ cm}^2}$. $P = 9 + 12 + 15 = \boxed{36 \text{ cm}}$.

29. ★★☆

a) $h = 96 : 12 = \boxed{8 \text{ cm}}$

b) $b \cdot 9 = 90 \Rightarrow b = \boxed{10 \text{ m}}$

30. ★★☆ Base 8 cm (sobre la línia de baix) i altura 5 cm: $A = \frac{8 \cdot 5}{2} = \boxed{20 \text{ cm}^2}$.

31. ★★★ $A = \frac{14 \cdot 12}{2} = \boxed{84 \text{ cm}^2}$. Amb base 15: $\frac{15 \cdot h}{2} = 84 \Rightarrow 15 \cdot h = 168 \Rightarrow h = \boxed{11{,}2 \text{ cm}}$.

32. ★★★ Tots tenen la mateixa base (10 cm) i la mateixa altura (6 cm, la distància entre les paral·leles): $A = \frac{10 \cdot 6}{2} = \boxed{30 \text{ cm}^2}$ $\boxed{\text{tots tres}}$.

Llicència Creative Commons Reconeixement-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

5 Àrea del rombe i del trapezi

33. ★☆☆

a) $\frac{16\cdot 10}{2} = \boxed{80\ \text{cm}^2}$ **b)** $\frac{9\cdot 6}{2} = \boxed{27\ \text{m}^2}$ **c)** $\frac{7{,}5\cdot 4}{2} = \boxed{15\ \text{cm}^2}$

34. ★☆☆

a) $\frac{(12+8)\cdot 5}{2} = \boxed{50\ \text{cm}^2}$ **b)** $\frac{(9+4)\cdot 6}{2} = \boxed{39\ \text{m}^2}$ **c)** $\frac{17\cdot 4}{2} = \boxed{34\ \text{cm}^2}$

35. ★★☆ $l = 52 : 4 = \boxed{13\ \text{cm}}$; $A = \frac{24\cdot 10}{2} = \boxed{120\ \text{cm}^2}$

36. ★★☆ $A = \frac{(18+10)\cdot 3}{2} = \boxed{42\ \text{cm}^2}$; $P = 18 + 10 + 5 + 5 = \boxed{38\ \text{cm}}$

37. ★★☆

a) $15 \cdot d = 120 \Rightarrow d = \boxed{8\ \text{cm}}$

b) $(14 + 10) \cdot h = 144 \Rightarrow 24 \cdot h = 144 \Rightarrow h = \boxed{6\ \text{m}}$

38. ★★★ $A = \frac{1{,}2\cdot 0{,}8}{2} = \boxed{0{,}48\ \text{m}^2} = \boxed{4800\ \text{cm}^2}$

39. ★★★ $A = \frac{(85+55)\cdot 40}{2} = 2800\ \text{m}^2 = \boxed{0{,}28\ \text{ha}}$. Preu: $2800 \cdot 12 = \boxed{33\,600\ €}$.

6 Àrea dels polígons regulars i dels polígons qualssevol

40. ★☆☆ $P = 5 \cdot 10 = \boxed{50\ \text{cm}}$; $A = \frac{50\cdot 6{,}88}{2} = \boxed{172\ \text{cm}^2}$

41. ★☆☆ $P = 6 \cdot 8 = \boxed{48\ \text{cm}}$; $A = \frac{48\cdot 6{,}93}{2} = \boxed{166{,}32\ \text{cm}^2}$

42. ★★☆ $l = 40 : 8 = \boxed{5\ \text{cm}}$; $A = \frac{40\cdot 6{,}04}{2} = \boxed{120{,}8\ \text{cm}^2}$

43. ★★☆ $\frac{36\cdot a}{2} = 93{,}6 \Rightarrow 36 \cdot a = 187{,}2 \Rightarrow a = \boxed{5{,}2\ \text{cm}}$

44. ★★☆ Rectangle 6×4 i triangle de base 6 i altura 2: $24 + \frac{6\cdot 2}{2} = 24 + 6 = \boxed{30\ \text{cm}^2}$

45. ★★★ Rectangle $7 \times 6 = 42$. Triangles que sobren: $\frac{7\cdot 2}{2} = 7$, $\frac{4\cdot 4}{2} = 8$ i $\frac{3\cdot 6}{2} = 9$. $A = 42 - 7 - 8 - 9 = \boxed{18\ \text{cm}^2}$

46. ★★★ Rajola: $A = \frac{60\cdot 8{,}66}{2} = 259{,}8\ \text{cm}^2$. $25\ \text{m}^2 = 250\,000\ \text{cm}^2$; $250\,000 : 259{,}8 \approx 962{,}3$: calen $\boxed{963\ \text{rajoles}}$.

Llicència Creative Commons Reconeixement-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

7 La circumferència i la seua longitud

47. ★☆☆

a) a: radi **b)** b: diàmetre **c)** c: corda **d)** d: arc

48. ★☆☆

a) $2 \cdot 3{,}14 \cdot 4 = \boxed{25{,}12 \text{ cm}}$ **b)** $2 \cdot 3{,}14 \cdot 9 = \boxed{56{,}52 \text{ m}}$

c) $3{,}14 \cdot 20 = \boxed{62{,}8 \text{ cm}}$ **d)** $3{,}14 \cdot 7 = \boxed{21{,}98 \text{ mm}}$

49. ★★☆

a) $d = 31{,}4 : 3{,}14 = 10 \text{ cm}$; $r = \boxed{5 \text{ cm}}$

b) $d = 75{,}36 : 3{,}14 = \boxed{24 \text{ m}}$; $r = \boxed{12 \text{ m}}$

50. ★★☆ $L = 3{,}14 \cdot 70 = \boxed{219{,}8 \text{ cm}}$. $1 \text{ km} = 100\,000 \text{ cm}$; $100\,000 : 219{,}8 \approx 455$ $\boxed{\text{unes 455 voltes}}$.

51. ★★☆

a) $\frac{2 \cdot 3{,}14 \cdot 8 \cdot 90}{360} = \boxed{12{,}56 \text{ cm}}$ **b)** $\frac{2 \cdot 3{,}14 \cdot 12 \cdot 60}{360} = \boxed{12{,}56 \text{ cm}}$ **c)** $\frac{2 \cdot 3{,}14 \cdot 6 \cdot 150}{360} = \boxed{15{,}7 \text{ cm}}$

52. ★★★

a) $\frac{3{,}14 \cdot 10}{2} + 10 = 15{,}7 + 10 = \boxed{25{,}7 \text{ cm}}$ **b)** $\frac{2 \cdot 3{,}14 \cdot 4}{4} + 4 + 4 = 6{,}28 + 8 = \boxed{14{,}28 \text{ cm}}$

53. ★★★ En 20 minuts gira $120°$: $\frac{2 \cdot 3{,}14 \cdot 15 \cdot 120}{360} = \boxed{31{,}4 \text{ cm}}$. En una hora: $2 \cdot 3{,}14 \cdot 15 = \boxed{94{,}2 \text{ cm}}$.

8 L'àrea del cercle i de les figures circulars

54. ★☆☆

a) $3{,}14 \cdot 36 = \boxed{113{,}04 \text{ cm}^2}$ **b)** $3{,}14 \cdot 6{,}25 = \boxed{19{,}625 \text{ m}^2}$

c) $3{,}14 \cdot 7^2 = \boxed{153{,}86 \text{ cm}^2}$ **d)** $3{,}14 \cdot 15^2 = \boxed{706{,}5 \text{ cm}^2}$

55. ★★☆

a) $\frac{3{,}14 \cdot 100}{2} = \boxed{157 \text{ cm}^2}$ **b)** $\frac{3{,}14 \cdot 36}{4} = \boxed{28{,}26 \text{ cm}^2}$

56. ★★☆

a) $\frac{3{,}14 \cdot 64 \cdot 45}{360} = \boxed{25{,}12 \text{ cm}^2}$ **b)** $\frac{3{,}14 \cdot 81 \cdot 120}{360} = \boxed{84{,}78 \text{ cm}^2}$ **c)** $\frac{3{,}14 \cdot 16 \cdot 270}{360} = \boxed{37{,}68 \text{ cm}^2}$

Llicència Creative Commons Reconeixement-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

57. ★★☆

a) $3{,}14 \cdot (49 - 16) = 3{,}14 \cdot 33 = \boxed{103{,}62 \text{ cm}^2}$ **b)** $3{,}14 \cdot (100 - 36) = 3{,}14 \cdot 64 = \boxed{200{,}96 \text{ m}^2}$

58. ★★☆

a) $r^2 = 78{,}5 : 3{,}14 = 25 \Rightarrow r = \boxed{5 \text{ cm}}$

b) $r = 18{,}84 : 6{,}28 = 3$ cm; $A = 3{,}14 \cdot 9 = \boxed{28{,}26 \text{ cm}^2}$

59. ★★★ Corona de radis $R = 6$ cm i $r = 0{,}75$ cm: $3{,}14 \cdot (36 - 0{,}5625) \approx \boxed{111{,}27 \text{ cm}^2}$

60. ★★★ Gran: $3{,}14 \cdot 16^2 = 803{,}84$ cm^2. Dues menudes: $2 \cdot 3{,}14 \cdot 11^2 = 759{,}88$ cm^2. La pizza gran.

9 Figures compostes

61. ★☆☆ Completem el rectangle 10×7 i restem el tros 4×3: $A = 70 - 12 = \boxed{58 \text{ cm}^2}$. $P = 10 + 4 + 4 + 3 + 6 + 7 = \boxed{34 \text{ cm}}$.

62. ★☆☆ $A = 8 \cdot 5 + \frac{8 \cdot 3}{2} = 40 + 12 = \boxed{52 \text{ m}^2}$

63. ★★☆ Semicercle de radi 2 cm: $A = 6 \cdot 4 + \frac{3{,}14 \cdot 2^2}{2} = 24 + 6{,}28 = \boxed{30{,}28 \text{ cm}^2}$. $P = 6 + 6 + 4 + \frac{2 \cdot 3{,}14 \cdot 2}{2} = 16 + 6{,}28 = \boxed{22{,}28 \text{ cm}}$

64. ★★☆ Els quatre quarts formen un cercle sencer de radi 4 cm: $A = 8^2 - 3{,}14 \cdot 4^2 = 64 - 50{,}24 = \boxed{13{,}76 \text{ cm}^2}$

65. ★★☆ Rectangle exterior: $(10 + 3) \times (6 + 3) = 13 \cdot 9 = 117$ m^2. $A = 117 - 60 = \boxed{57 \text{ m}^2}$

66. ★★☆ Els dos semicercles fan un cercle de radi 25 m: $A = 80 \cdot 50 + 3{,}14 \cdot 25^2 = 4000 + 1962{,}5 = \boxed{5962{,}5 \text{ m}^2}$. $P = 2 \cdot 80 + 3{,}14 \cdot 50 = 160 + 157 = \boxed{317 \text{ m}}$

67. ★★★ Centre: $3{,}14 \cdot 9 = \boxed{28{,}26 \text{ cm}^2}$. Zona del mig: $3{,}14 \cdot (36 - 9) = \boxed{84{,}78 \text{ cm}^2}$. Zona de fora: $3{,}14 \cdot (81 - 36) = \boxed{141{,}3 \text{ cm}^2}$.

68. ★★★ La diagonal divideix la fulla en dues meitats. Cada meitat és un quart de cercle menys un triangle: $\frac{3{,}14 \cdot 36}{4} - \frac{6 \cdot 6}{2} = 28{,}26 - 18 = 10{,}26$. $A = 2 \cdot 10{,}26 = \boxed{20{,}52 \text{ cm}^2}$

69. ★★★ $A = 1{,}2 \cdot 1{,}5 + \frac{3{,}14 \cdot 0{,}6^2}{2} = 1{,}8 + 0{,}5652 = 2{,}3652$ m^2. $2{,}3652 \cdot 45 \approx \boxed{106{,}43 \text{ €}}$

Llicència Creative Commons Reconeixement-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

10 El teorema de Pitàgores

70. ★☆☆

a) Catets x i y; hipotenusa z: $\boxed{z^2 = x^2 + y^2}$

b) Catets n i p; hipotenusa m: $\boxed{m^2 = n^2 + p^2}$

c) Catets r i s; hipotenusa t: $\boxed{t^2 = r^2 + s^2}$

71. ★☆☆

a) $\sqrt{9+16} = \sqrt{25} = \boxed{5 \text{ cm}}$

b) $\sqrt{36+64} = \sqrt{100} = \boxed{10 \text{ m}}$

c) $\sqrt{25+144} = \sqrt{169} = \boxed{13 \text{ cm}}$

d) $\sqrt{81+144} = \sqrt{225} = \boxed{15 \text{ dm}}$

72. ★☆☆

a) $\sqrt{100-36} = \sqrt{64} = \boxed{8 \text{ cm}}$

b) $\sqrt{169-144} = \sqrt{25} = \boxed{5 \text{ m}}$

c) $\sqrt{289-225} = \sqrt{64} = \boxed{8 \text{ cm}}$

d) $\sqrt{625-49} = \sqrt{576} = \boxed{24 \text{ cm}}$

73. ★★☆

a) $c = \sqrt{4+25} = \sqrt{29} \approx \boxed{5{,}39 \text{ cm}}$

b) $c = \sqrt{16+36} = \sqrt{52} \approx \boxed{7{,}21 \text{ cm}}$

c) $b = \sqrt{64-9} = \sqrt{55} \approx \boxed{7{,}42 \text{ cm}}$

d) $b = \sqrt{100-49} = \sqrt{51} \approx \boxed{7{,}14 \text{ cm}}$

74. ★★☆

a) $10^2 = 100$; $36 + 64 = 100$: $\boxed{\text{sí}}$

b) $9^2 = 81$; $25 + 49 = 74$: $\boxed{\text{no}}$

c) $17^2 = 289$; $64 + 225 = 289$: $\boxed{\text{sí}}$

d) $15^2 = 225$; $100 + 121 = 221$: $\boxed{\text{no}}$

75. ★★☆ L'escala és la hipotenusa: $h = \sqrt{5^2 - 1{,}4^2} = \sqrt{25 - 1{,}96} = \sqrt{23{,}04} = \boxed{4{,}8 \text{ m}}$

76. ★★☆ Els dos trajectes són els catets: $d = \sqrt{12^2 + 9^2} = \sqrt{144 + 81} = \sqrt{225} = \boxed{15 \text{ km}}$

77. ★★★ Triangle rectangle de catets 6 cm i 8 cm: $AB = \sqrt{36 + 64} = \sqrt{100} = \boxed{10 \text{ cm}}$

78. ★★★ Cada cable: $\sqrt{12^2 + 5^2} = \sqrt{169} = 13$ m. Quatre cables: $4 \cdot 13 = \boxed{52 \text{ m}}$

Llicència Creative Commons Reconeixement-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

11 Pitàgores per a calcular àrees i perímetres

79. ★☆☆

a) $\sqrt{64+36} = \boxed{10\text{ cm}}$ **b)** $\sqrt{576+49} = \boxed{25\text{ m}}$ **c)** $\sqrt{16+16} = \sqrt{32} \approx \boxed{5{,}66\text{ cm}}$

80. ★☆☆ L'altura parteix la base en dues meitats de 8 cm: $h = \sqrt{10^2-8^2} = \sqrt{36} = \boxed{6\text{ cm}}$. $A = \frac{16\cdot 6}{2} = \boxed{48\text{ cm}^2}$. $P = 16+10+10 = \boxed{36\text{ cm}}$

81. ★★☆ Les semidiagonals fan 9 i 12 cm: $l = \sqrt{81+144} = \sqrt{225} = \boxed{15\text{ cm}}$. $P = 4\cdot 15 = \boxed{60\text{ cm}}$. $A = \frac{18\cdot 24}{2} = \boxed{216\text{ cm}^2}$

82. ★★☆ Triangle de catets $12-7=5$ i 12: $c = \sqrt{25+144} = \sqrt{169} = 13$ cm. $P = 12+7+12+13 = \boxed{44\text{ cm}}$. $A = \frac{(12+7)\cdot 12}{2} = \boxed{114\text{ cm}^2}$

83. ★★☆ A cada costat sobren $(16-4):2 = 6$ cm: $h = \sqrt{100-36} = \boxed{8\text{ cm}}$. $A = \frac{(16+4)\cdot 8}{2} = \boxed{80\text{ cm}^2}$. $P = 16+4+10+10 = \boxed{40\text{ cm}}$

84. ★★☆ El radi és igual al costat (4 cm): $a = \sqrt{4^2-2^2} = \sqrt{12} \approx \boxed{3{,}46\text{ cm}}$. $P = 6\cdot 4 = 24$ cm; $A = \frac{24\cdot 3{,}46}{2} = \boxed{41{,}52\text{ cm}^2}$

85. ★★★ Costat: $18:3 = 6$ cm. $h = \sqrt{6^2-3^2} = \sqrt{27} \approx \boxed{5{,}20\text{ cm}}$. $A = \frac{6\cdot 5{,}2}{2} = \boxed{15{,}6\text{ cm}^2}$

86. ★★★ La diagonal del quadrat és un diàmetre (8 cm): $l^2+l^2 = 64 \Rightarrow l^2 = 32$, que és l'àrea del quadrat. $A = 3{,}14\cdot 4^2 - 32 = 50{,}24 - 32 = \boxed{18{,}24\text{ cm}^2}$

87. ★★★ Cada costat de la teulada: $\sqrt{4^2+3^2} = \sqrt{25} = 5$ m. $P = 8+5+5+5+5 = \boxed{28\text{ m}}$. $A = 8\cdot 5 + \frac{8\cdot 3}{2} = \boxed{52\text{ m}^2}$

Llicència Creative Commons Reconeixement-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)